

Exploring the Key Factors Affecting Sales in Ecuador's Favorita Supermarkets

1.Introduction

The retail landscape in Ecuador is shaped by complex internal and external factors influencing consumer behavior and sales performance at major chains like Favorita. As the retail industry evolves, supermarkets must adapt to dynamic market conditions, including changing consumer preferences, economic shifts, and competitive pressures. This project seeks to explore the key factors affecting sales in Favorita supermarkets by analyzing a rich set of data that includes both internal store-specific variables and external socio-economic influences.

The primary dataset, sourced from [Kaggle](#), offers detailed information of sales data across different store-product family combinations, including temporal aspects like promotions, holidays, and events. The complementary datasets include external data such as oil prices and economic events, which provides a broader view of the socio-economic factors affecting consumer spending. By identifying key drivers of sales performance, this paper could help the supermarket chain optimize operations and deliver a more tailored and effective customer experience, ultimately driving greater profitability and market share.

2.Data and Methodology

2.1 Data Description

The primary dataset used in this analysis consists of historical sales data from Favorita supermarkets across Ecuador, covering the period from January 5, 2013, to August 15, 2017 (however, only data from periods prior to April 15, 2016, is included in the analysis due to the earthquake that occurred on April 16, 2016). It includes key variables such as `store_nbr` (store number), `family` (product family category), `date` (transaction date), `sales` (sales amount for each store-product-family combination), and `transactions` (number of transactions per store-product-family combination on a given day).

Supplementary datasets that enhance the analysis are included: (1) The Holidays and Events Dataset (`holidays_events.csv`) provides details on national holidays and special events that may influence consumer behavior, with columns including `date`, `holiday` (name of the holiday or event), and `holiday_type` (e.g., national, regional). (2) The Oil Prices Dataset (`oil.csv`) tracks daily crude oil prices in Ecuador, a critical factor given the country's oil-dependent economy, with fields `date` and `dcoilwtico` (daily closing price). (3) The Stores Dataset (`stores.csv`) offers metadata for each store, including `store_nbr`, `store_type` (e.g., urban, rural), and location details (city, state). (4) The Transactions Dataset (`transactions.csv`) records daily transaction counts per store, providing insight into store-level activity. Together, these datasets enable a comprehensive examination of sales patterns and their influencing factors.

Moreover, in Ecuador, public sector wages are disbursed biweekly, on the 15th and last day of each month, potentially increasing disposable income and driving supermarket sales on these payday. This influence of wage cycles should also be addressed in the models to analyze sales.

The overall sales pattern exhibits seasonal fluctuations, with notable peaks around holidays and paydays, illustrated by Figure 1 below. This suggests strong seasonality which would be considered in the models.

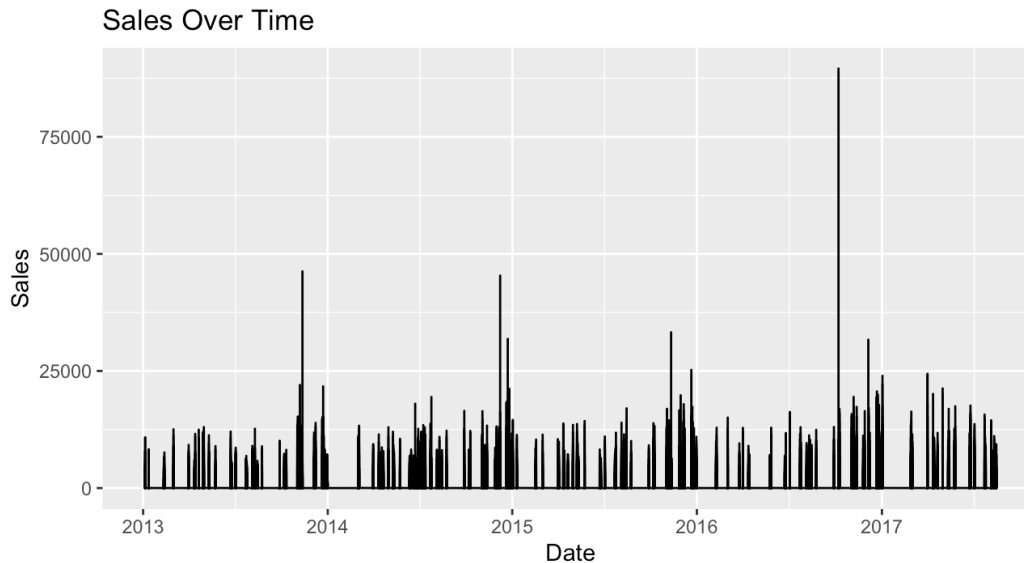


Figure 1: Total Sales in Ecuador's Favorita Supermarkets Over Time

2.2 Data Pre-Processing

To prepare the datasets for analysis, the four external datasets were merged onto the sales dataset by dates. Missing values in the oil price column (common on weekends and holidays) were imputed by propagating the last observed value forward to maintain a continuous time series, as oil prices today is likely to be similar to prices yesterday without external shocks. Additionally, to minimize the impact of the magnitude 7.8 earthquake that occurred in Ecuador on April 16, 2016, sales records are filtered to include only data from periods before April 15, 2016.

Dummies were created for key categorical and temporal variables for the subsequent regression models. These include `holiday_dummy` to denote holiday periods, `payday_dummy` to identify biweekly wage disbursement dates, and month dummies to account for seasonal effects. The `payday_dummy` spans an eight-day window, covering days 14-17 and 28-31 of each month, to capture the potential spillover effects of biweekly paydays (on the 15th and last day of the month). This period reflects the immediate impact of wage payments, assuming consumers may spend shortly after payday, influencing sales patterns. Furthermore, five `payday_dummy` lags were introduced to analyze delayed effects of consumer spending, especially for goods that are durable. These dummy variables help capture the longer-term influence of wage disbursement and external events on sales behavior.

Data Analysis

2.3.1 Multiple Linear Regression:

To start with, a multiple linear regression model with fixed effects is implemented to analyze the key factors influencing sales. This model assesses the impact of promotions,

holidays, and paydays while controlling for store characteristics and economic trends.

The estimated model is:

$$Sales_{ist} = \beta_0 + \beta_1 \cdot onpromotion_{ist} + \beta_2 \cdot dcoilwtico_t + \beta_3 \cdot holidaydummy_t + \beta_4 \cdot paydaydummy_t + \sum_{m=1}^{12} month_{m,t} + \alpha_{store_i} + \alpha_{family_j} + \alpha_{city_k} + \alpha_{state_l} + \varepsilon_{ist}$$

where:

i : indexes the store,

s : product family,

t : time (date),

α_{store_i} : store fixed effect

α_{family_j} : product family fixed effect

α_{city_k} : city fixed effect

α_{state_l} : state fixed effect

ε_{ist} : error term

This model would provide us with an initial understanding of the factors influencing sales, which later models could be built on.

2.3.2 Regularized Regression Analysis

To identify key determinants of sales and mitigate potential multicollinearity among predictors, regularized regression methods are employed, with four LASSO models and an elastic net model. LASSO applies an L1 penalty to regression coefficients, shrinking less influential coefficients to zero, which facilitates variable selection and enhances model interpretability. This approach is particularly effective for high-dimensional datasets with correlated predictors, as it isolates the most impactful variables while reducing overfitting. The elastic net model combines L1 and L2 penalties, making it useful in cases of highly correlated predictors.

The following regularized regression models are estimated using different sets of interactions and fixed effects to determine the most predictive factors:

1. Baseline Lasso Model (Without Interactions)

This initial model includes all the predictors from the fixed effect models but adds payday_dummy lags (up to 5 periods) which acknowledges the extended impact of income availability on consumer purchasing patterns, as spending behavior often stretches beyond the immediate payday. It aims to capture the change of purchasing power throughout pay cycles.

Categorical variables are re-leveled to the most frequent categories for reference and interpretation. The most frequent category for city, product family, month, and state are Quito (which is the capital of Ecuador), Automotive, 12 (December), and Pichincha (the state that Quito is located in), respectively. This re-leveling approach enables more precise analysis of regional and seasonal trends by comparing the results to the base level category, such as identifying how sales in other cities deviate from Quito.

2.Lasso with Many Interactions

The second model builds on the first lasso model by adding interaction terms between store ID, product family, promotions, and time-related variables to capture heterogeneity across stores and products. The reason behind is that the impact of promotions (onpromotion) is unlikely to be uniform across all stores and product families. Certain stores might experience stronger promotional effects due to local demand, competition, or customer demographics. Similarly, some product categories may be more responsive to discounts than others. For instance, grocery products might exhibit a stronger sales response to promotions compared to luxury items. By including interaction terms between store ID, product family, and promotion, the model can capture these nuanced differences in consumer behavior to better understand how these factors collectively impact sales.

3.Lasso with Lag Interactions

The third model enhances the previous lasso model by introducing temporal dynamics related to payday effects. While the second model captured interactions between promotions, stores, and product families, this model specifically focuses on how payday impacts extend over time by including interactions of lags of `payday_dummy` (up to five periods) with other factors. These lag interactions allow the model to detect patterns such as initial large purchases immediately following payday, followed by smaller, possibly more discretionary purchases in subsequent weeks. The five-period lag structure provides sufficient range to model the complete cycle of payday-influenced consumer behavior, capturing both immediate effects and delayed spending decisions.

4.Lasso with Seasonality Trends

The fourth model implements a sophisticated time-series decomposition approach, explicitly modeling both long-term trends and cyclical patterns, aiming to capture the strong and complicated seasonal trend from the total sales trend graph (Fig. 1). It incorporates polynomial time terms (linear, quadratic, and cubic) with the other predictors in the fixed effect model to capture non-linear growth or decline in sales over the observation period, addressing potential market expansion and/or evolution patterns.

5.Elastic Net Model

The fifth model uses an elastic net implementation build on the predictors from the first lasso model that combines L1 and L2 penalties ($\alpha=0.5$) to address the correlation among predictors. In retail sales data, where predictors like neighboring stores or complementary product categories often exhibit correlation, this elastic net model provides more stable and reliable coefficient estimates.

2.3.3 Time Series Analysis

The time series analysis implements an autoregressive model with exogenous variables to capture the temporal dynamics of aggregate sales across all stores and products. By aggregating data to the daily level and incorporating a log transformation of the response variable, this model addresses both the high variability in sales volumes and the temporal dependencies in the data.

The estimated model is:

$$\begin{aligned}\log(\text{sales}_t) = & \beta_0 + \beta_1 \log(\text{sales}_{t-1}) + \beta_2 \log(\text{sales}_{t-2}) + \beta_3 \log(\text{sales}_{t-7}) + \beta_4 t \\ & + \beta_5 \text{month}_t + \beta_6 \text{avg_onpromotion}_t + \beta_7 \text{avg_dcoilwtico}_t \\ & + \beta_8 \text{payday_dummy}_t + \beta_9 \text{lag_payday_dummy}_{t-1} \\ & + \beta_{10} \text{lag_payday_dummy}_{t-2} + \beta_{11} \text{lag_payday_dummy}_{t-3} + \epsilon_t\end{aligned}$$

where:

t : time

avg_dcoilwtico_t : average oil price at time t

ϵ_t : error term

Specifically, the log transformation of total sales stabilizes variance across the time series, addressing the heteroscedasticity often present in retail data where sales volatility tends to increase with volume. The inclusion of autoregressive components at lags 1, 2, and 7 through the lagged dependent variables recognizes multiple temporal dependencies in the sales patterns.

The lag 1 and lag 2 terms capture short-term momentum in consumer purchasing behavior, where today's sales are influenced by recent performance. These components serve as proxies for unobserved persistent factors including customer loyalty and retailer reputation effects. Meanwhile, the lag 7 term addresses the weekly cyclical pattern evident in the sales data, capturing day-of-week effects where consumer shopping behaviors follow predictable weekly patterns, perhaps based on payday schedules, weekend availability and routine.

Beyond the autoregressive structure, the model incorporates calendar-based predictors of holiday and payday indicators to capture recurring but irregular demand shocks. These binary variables isolate the impact of significant consumer spending events, distinguishing them from underlying sales trends. The model further includes a continuous time trend variable to capture long-term market evolution.

2.3.4 K-means Clustering

To further enhance the analysis of sales patterns, K-means clustering is employed to segment stores based on their geographic and sales characteristics, providing a complementary perspective to the regression model. The log-transformed total sales serve as a key input feature for clustering. By incorporating this transformed sales data alongside geographic variables of state and city, K-means identifies natural groupings of stores that exhibit similar sales behavior and regional traits.

The algorithm iteratively assigns stores to clusters by minimizing the within-cluster variance, using Euclidean distance to measure similarity across these features. The application of K-Means clustering is helpful for analyzing sales data because it uncovers spatial and operational patterns that the regression model alone might overlook. For example, the autoregressive component captures temporal dependencies like consumer purchasing inertia, yet they do not explicitly account for spatial heterogeneity across stores. K-Means addresses this gap by grouping stores with similar sales dynamics and

geographic contexts. By identifying such groupings, businesses can tailor strategies like promotional campaigns to specific store clusters, optimizing resource allocation.

3.Results

3.1.1 Multiple Linear Regression Results

Dependent Var.:	sales
onpromotion	18.80*** (1.675)
dcoilwtico	-1.884*** (0.2191)
payday_dummy	-9.535. (5.101)
month1	-238.1*** (20.90)
month2	-166.3*** (14.23)
month3	-79.45*** (8.847)
month4	-147.1*** (14.69)
month5	-116.4*** (12.07)
month6	-130.4*** (14.81)
month7	-80.30*** (11.58)
month8	-138.5*** (15.15)
month9	-89.34*** (10.18)
month10	-113.5*** (13.51)
month11	-123.6*** (9.980)
Fixed-Effects:	-----
store_nbr	Yes
family	Yes
city	Yes
state	Yes
S.E.: Clustered	by: store_nbr
Observations	436,590
R2	0.57142
Within R2	0.08485

Signif. codes:	0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Table 1: Fixed Effects Panel Regression Results for Store-Level Sales

From the above results, we could see that promotional activities significantly boost sales while oil price increases negatively impact purchasing. There is a strong seasonal patterns shown by December outperforming all other months, and January experiencing the steepest decline. The model explains 57% of sales variation suggested by its R^2 , with store-product fixed effects accounting for most explanatory power. The within R^2 of 0.08485 indicates that the time-varying predictors explain about 8.5% of the variation within each store-product combination. This model gives us an overall understanding of the data, and the following models would continue to build on this with more details.

3.1.2 Regularized Regression Results

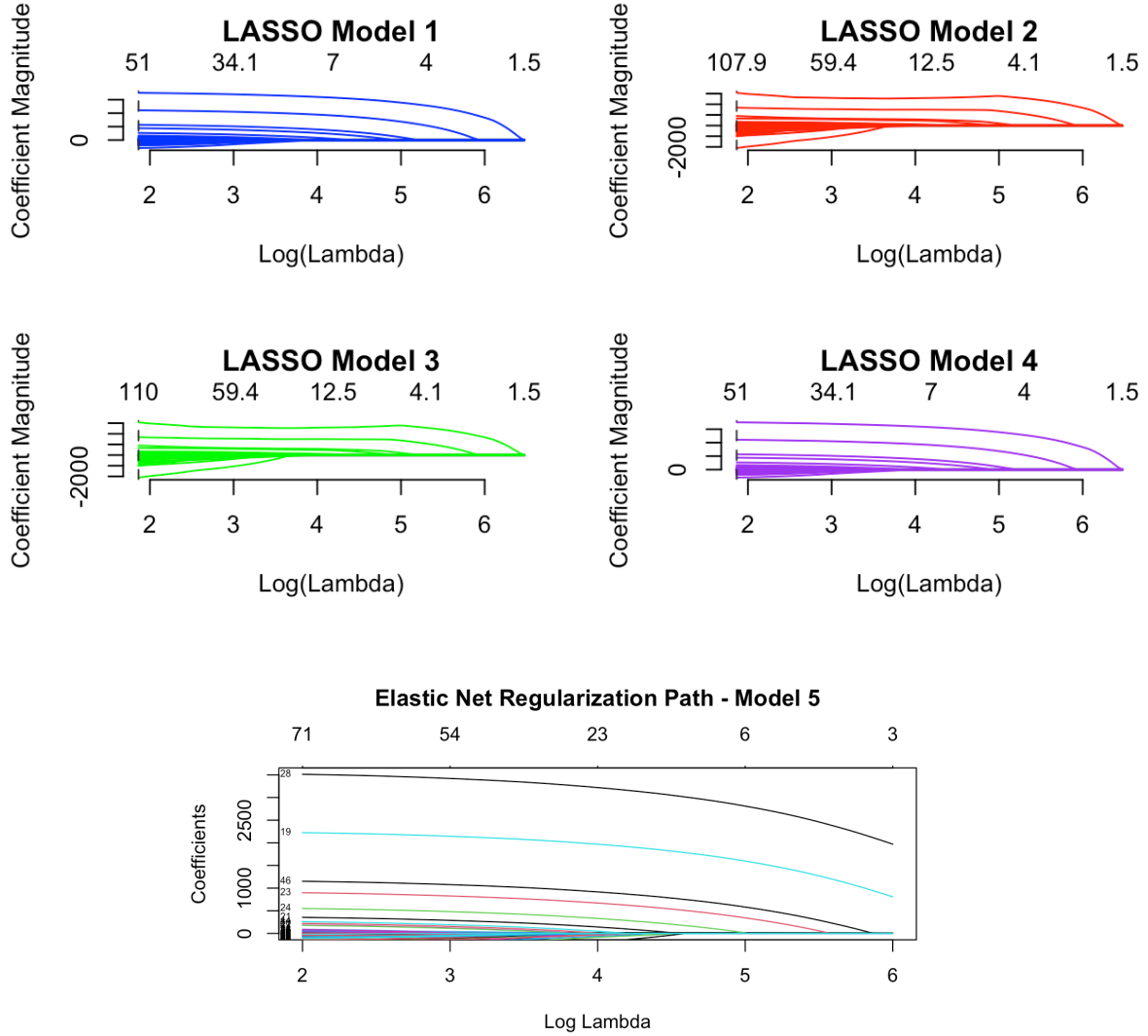


Figure 2: Coefficient Regularization Paths for LASSO and Elastic Net Models

To ensure robust model selection and prevent overfitting, we implemented 10-fold cross-validation for each of our models. Cross-validation helps optimize the regularization parameter (λ) by splitting the dataset into ten subsets, iteratively training the model on nine subsets while validating on the remaining one. This process is repeated ten times, and the final λ is chosen based on mean cross-validation error.

For selecting this specific final λ to use on the last testing set of data, we used the one-standard-error rule, which selects a regularization parameter that is within one standard error of the optimal λ . This choice favors simpler models with fewer non-zero coefficients. It makes the results more interpretable and reducing the risk of overfitting, aligning with my intention on causal inference instead of prediction as to determine the factors that influence sales. Each of the five models are applied with this cross-validation process to determine the lambda.

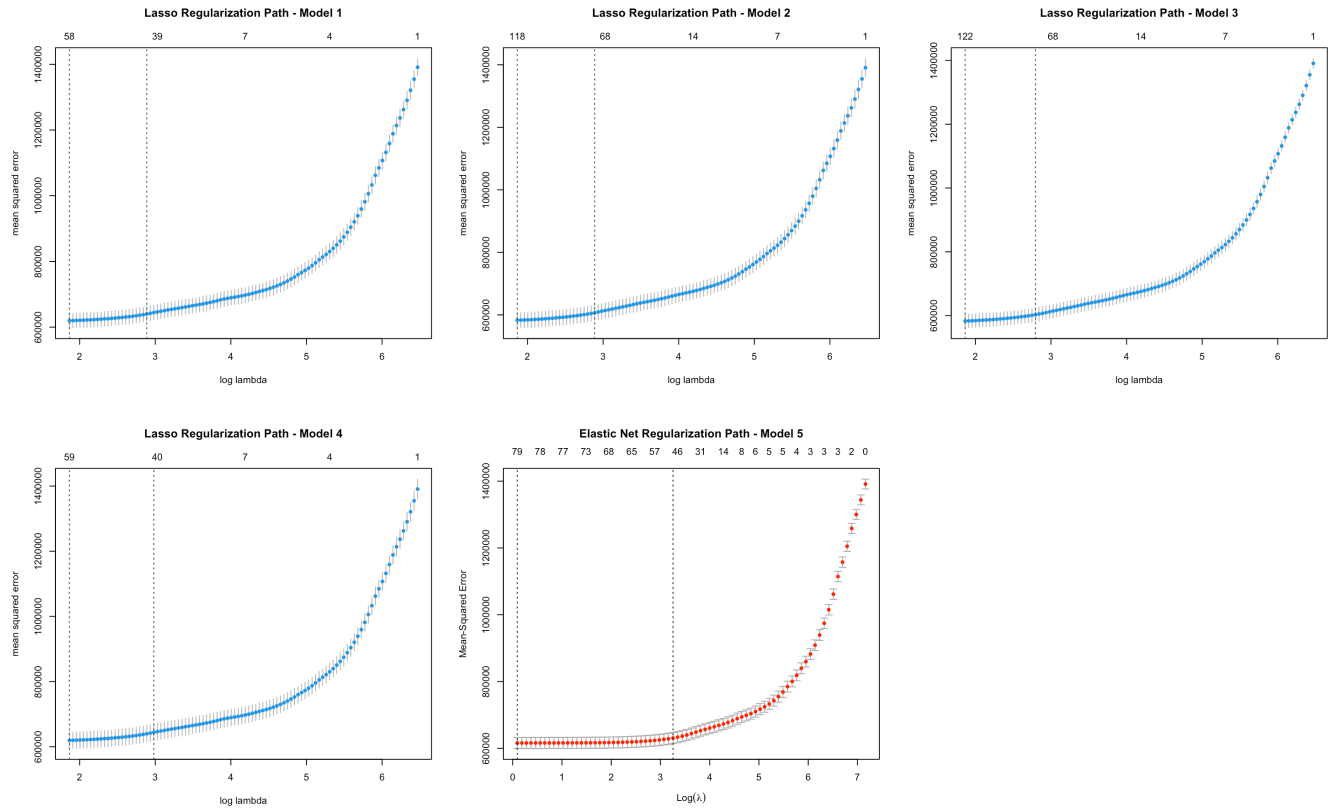


Figure 3: Cross-Validation Error Regularization Paths for LASSO and Elastic Net Models

The results of these cross-validation regularization paths for the five models, as shown in the above graphs, illustrate the relationship between $\log(\lambda)$ and mean squared error, with Models 1-4 (Lasso) displaying a steep error increase as $\log(\lambda)$ rises, indicating rapid simplification as coefficients are forced to zero, while Model 5 of the elastic net model shows a slower error rise due to its balanced L1 and L2 penalties as expected.

After choosing the optimal λ values by the one-standard-error rule, the corrected Akaike Information Criterion (AICc) is calculated at these points to compare these models. The AICc balances model fit and complexity by penalizing excessive parameters to mitigate overfitting, providing a robust metric for comparing models. For each of the five models, the AICc is computed at its respective λ_{1se} and compared.

Model	AICc Value
Model 1	5,837,517
Model 2	5,811,209
Model 3	5,809,786
Model 4	5,831,590
Elastic Net	5,832,182

Table 2: AICc Values Comparison Across Regularization Models

According to the AICc values computed, Model 3 has the lowest value and is thus considered as the best-performing model. This suggests that capturing the extended impact of payday-related income through interactions between payday_dummy and its

lagged variables is crucial for modeling consumer purchasing behavior. Its regularization path confirms this by showing a steep error increase at higher $\log(\lambda)$ values, indicating effective simplification at the chosen λ_{1se} , making it ideal for identifying key sales drivers while minimizing overfitting.

By rerunning model 3 on the entire data sample, the factors that most affected sales selected by model 3 are:

Category	Feature	Coefficient
Baseline	intercept	208.8329412
Location Effects	cityAmbato	-104.0753014
	cityCayambe	48.16802272
	cityCuenca	-301.5100925
	cityCuenca:stateAzuay	-66.28201762
	cityDaule	31.88571627
	cityDaule:stateGuayas	24.79267321
	cityEsmeraldas	-359.2922578
	cityGuaranda	-79.44652734
	cityGuaranda:stateBolivar	-119.6890591
	cityIbarra	-186.1040005
	cityIbarra:stateImbabura	-21.80061789
	cityLatacunga	-68.48428418
	cityLatacunga:stateCotopaxi	-138.0009763
	cityLibertad	-2.712508938
	cityLoja	-275.0659198
	cityMachala	-357.2617087
	cityManta	-26.48021298
	cityPlayas	-120.7888932
	cityPuyo	-394.1803343
	cityQuevedo	-69.37750717
	cityRiobamba	-179.6629514
	cityRiobamba:stateChimborazo	-9.87842533
	citySalinas	-251.1946397
	citySanto Domingo	-171.4214492
Economic Factors	dcoilwtico (oil price)	-1.756869962
Product Categories	familyBEVERAGES	1684.570614
	familyBREAD/BAKERY	224.5910029
	familyCLEANING	671.9434017
	familyDAIRY	354.1810223
	familyDELI	14.42562036
	familyEGGS	41.18071267
	familyFROZEN FOODS	59.56976063
	familyGROCERY I	3073.679419
	familyHOME CARE	43.15593054
	familyLIQUOR,WINE,BEER	19.43074343

	familyMEATS	117.3400153
	familyPERSONAL CARE	89.77465148
	familyPOULTRY	168.5173227
	familyPREPARED FOODS	10.00654802
	familyPRODUCE	901.5002197
Temporal Effects	month1	-65.18966956
	month1:familyBEVERAGES	-979.0311791
	month1:familyDAIRY	-8.997675986
	month1:familyGROCERY I	-2111.523357
	month1:familyPRODUCE	-731.2351813
	month10:familyGROCERY I	-587.1065063
	month11	-15.49446513
	month11:familyFROZEN FOODS	-22.39092214
	month11:familyGROCERY I	-512.2177966
	month11:familyPRODUCE	15.38868224
	month2	-16.42083479
	month2:familyBEVERAGES	-178.1041055
	month2:familyGROCERY I	-1012.449218
	month2:familyPRODUCE	-272.2371996
	month3:familyGROCERY I	-165.0156868
	month4	-4.72324692
	month4:familyBEVERAGES	-295.6714707
	month4:familyGROCERY I	-535.8404032
	month4:familyPRODUCE	-497.1538429
	month4:payday dummy	-35.33988989
	month5:familyBEVERAGES	-42.20181835
	month5:familyGROCERY I	-108.1691605
	month5:familyPRODUCE	-426.1224402
	month6	-0.874123149
	month6:familyBEVERAGES	-282.041082
	month6:familyCLEANING	-67.75302738
	month6:familyGROCERY I	-889.7404397
	month6:familyPRODUCE	-261.346781
	month7:familyBEVERAGES	108.4557298
	month7:familyGROCERY I	-570.862114
	month7:familyHOME CARE	4.75832131
	month7:familyPRODUCE	186.9672772
	month8	-18.14448493
	month8:familyBEVERAGES	-32.21896236
	month8:familyGROCERY I	-469.258519
	month8:familyPRODUCE	-95.30189205
	month9:familyGROCERY I	-340.0803207
Promotional Effects	onpromotion	19.78316285
	onpromotion:month1	44.54612447

	onpromotion:month11	-1.464612585
	onpromotion:month3	-0.272847611
	onpromotion:month4	-1.124183366
	onpromotion:month5	-5.393076883
	onpromotion:month6	1.997393768
	onpromotion:month8	-3.659965754
	onpromotion:month9	-6.079798363
State Effects	stateGuayas	-322.9652884
	stateLos Rios	-245.4217932
	stateManabi	-593.3370372
	stateSanto Domingo de los Tsachilas	-34.52640989
	stateTungurahua	-141.9000611
Store Effects	store_nbr	5.91499554
	store_nbr:familyBEVERAGES	22.69297872
	store_nbr:familyBREAD/BAKERY	3.998552958
	store_nbr:familyCLEANING	7.973363549
	store_nbr:familyDAIRY	6.600301358
	store_nbr:familyDELI	1.28694494
	store_nbr:familyEGGS	0.403180492
	store_nbr:familyFROZEN FOODS	3.912442592
	store_nbr:familyGROCERY I	33.6573887
	store_nbr:familyHOME CARE	0.966098849
	store_nbr:familyLIQUOR,WINE,BEER	0.620267577
	store_nbr:familyMEATS	2.982374433
	store_nbr:familyPERSONAL CARE	2.412831298
	store_nbr:familyPOULTRY	2.557646602
	store_nbr:familyPREPARED FOODS	0.360810244
	store_nbr:familyPRODUCE	13.69311389

Table 3: Significant Coefficients selected from Model 3 by Category

From the above data, regional disparities, global economic shocks, consumer preferences, and seasonality clearly shape sales outcomes. Compared to Quito, the baseline city, other regions exhibit significant variations. Ambato faces sales challenges due to weaker economic activity and infrastructure limitations, restricting retail growth. Esmeraldas, with its reliance on volatile oil and fishing industries, experiences even steeper declines, reflecting economic instability. In contrast, Daule benefits from its strong agricultural base, particularly rice production, which supports higher consumer spending. State-level differences further amplify these disparities, influenced by local governance, economic policies, and infrastructure investments.

Economic factors play a crucial role in sales trends. Rising oil prices (-1.76 per unit increase) exert financial pressure by raising costs and reducing disposable income. However, rather than causing an overall spending decline, consumer purchasing behavior shifts. Essential goods like groceries maintain strong demand, while beverages see a surge, possibly reflecting consumers' preference for affordable indulgences during

economic downturns. The dominant presence of the automotive category in Quito suggests that long-term investments, such as vehicles, remain resilient even amid oil price fluctuations.

Seasonality significantly influences purchasing patterns. A post-holiday slump in January and February contrasts with a December peak (most frequent month), most likely driven by holiday bonuses, festive spending, and perhaps promotional campaigns. Mid-year sales rebounds, such as the increase in beverage sales in July and produce sales, suggest seasonal income cycles or weather-driven demand shifts.

Promotions provide a notable sales boost, increasing sales by nearly 20 units on average. However, their effectiveness varies across time periods and product categories. While December promotions reinforce holiday-driven spending, off-season months experience weaker effects. Certain product categories, such as frozen foods and grocery staples, respond more sharply to seasonal demand shifts, especially compared to the base level of automotive which is less responsive to seasonality. This suggests that promotions should cater towards different products across different timings.

What is interesting about this model is that none of the payday variables (payday_dummy and its lags) are selected. This aligns with the permanent income hypothesis and consumption smoothing theory, which suggest that consumers do not significantly alter their spending patterns based solely on short-term income fluctuations. Instead, individuals tend to spread their consumption over time based on their lifetime wealth and income.

The deviance calculated for this model 3 is 57.18915 %, suggesting that this model explains 57.19% of the variation in Favorita's sales patterns. The results from this model underscores regional, economic, temporal, and promotional factors interact in dynamic ways to shape sales in Ecuador.

3.1.3 Time Series Analysis Results

As none of the regularized regression models selected the payday_dummy and its lags, perhaps due to the regularization penalizing variables that are highly correlated with other predictors, time series analysis is specifically employed to analyze the effect of time on sales particularly the persistence of sales trends over time. The inclusion of lagged sales variables in the time series model allows for capturing the autoregressive behavior of sales, acknowledging that past sales patterns often influence future sales outcomes. Unlike the regularized models, which are designed to prioritize simplicity and remove multicollinearity, the time series approach can better account for the temporal dependencies and capture both short-term dependencies (AR effects) and exogenous influences. The result is shown below:

Dependent variable:	
log_sales	
lag_log_sales_1	0.216*** (0.024)

lag_log_sales_2	-0.057** (0.023)
lag_log_sales_7	0.313*** (0.023)
time	0.007*** (0.001)
month2	0.019 (0.035)
month3	0.082** (0.034)
month4	0.032 (0.034)
month5	0.050 (0.034)
month6	0.081** (0.035)
month7	0.086** (0.035)
month8	0.032 (0.035)
month9	0.101*** (0.037)
month10	0.080** (0.036)
month11	0.100*** (0.036)
month12	0.220*** (0.038)
avg_onpromotion	0.005 (0.005)
avg_dcoilwtico	-0.001 (0.001)
payday_dummy	0.009 (0.023)
lag_payday_dummy_1	-0.036 (0.028)
lag_payday_dummy_2	0.006

	(0.028)
lag_payday_dummy_3	0.056** (0.023)
Constant	6.786*** (0.445)

Observations	1,676
Log Likelihood	-304.931
Akaike Inf. Crit.	653.863
=====	
Note:	*p<0.1; **p<0.05; ***p<0.01

Table 4: Results of Time Series Regression Analysis on Logarithm of Sales

The result of this model is similar to the results of the lasso model, suggesting how promotional effects, economic factors, and seasonal effects all influence sales. The `payday_dummy` is not statistically significant at 10%, but its third lag is significant, indicating a delayed effect of payday on sales. Overall, this model confirms the patterns found in the previous lasso model.

The ACF plot (Figure 4 shown below) displays minimal autocorrelation in the residuals across different lag periods. Apart from lag 0 (which naturally shows perfect correlation of 1.0), most of the other lags fall within the blue dashed confidence bands, indicating that they are not statistically significant. This suggests that the model with lag 1,2,7 autoregressive components has successfully captured most of the temporal patterns in the data.

Autocorrelation of Residuals

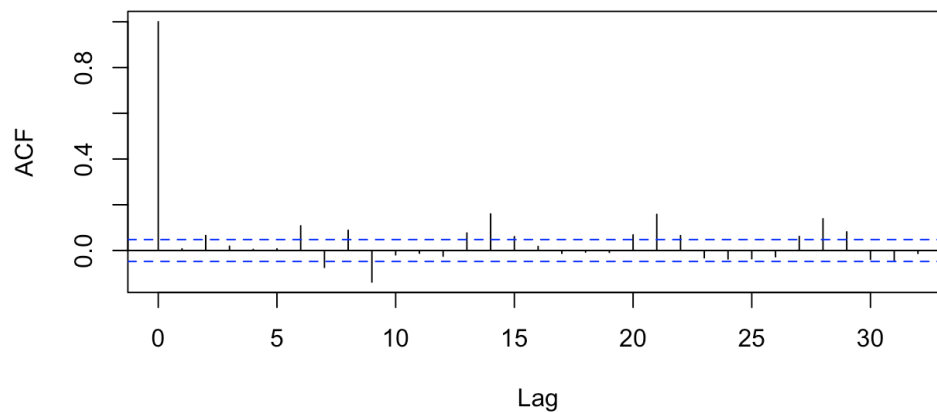


Figure 4: ACF on residuals of Time Series Regression Analysis on Logarithm of Sales

3.1.4 K-means Clustering Results

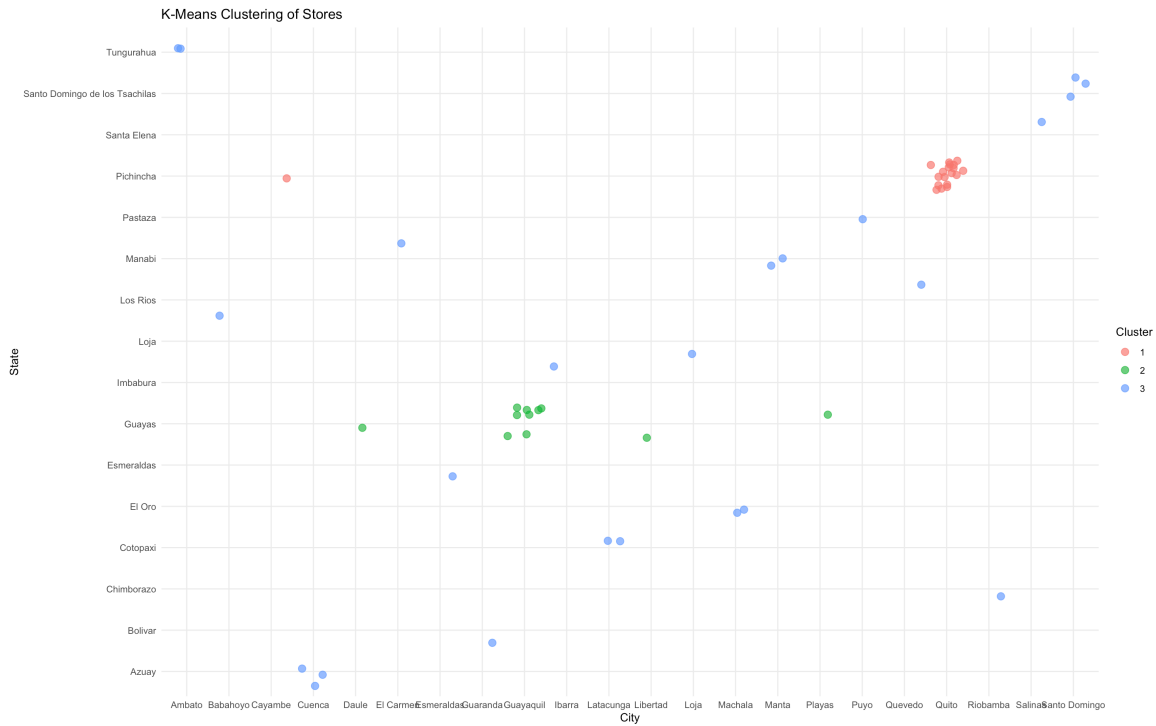


Figure 5: K-Means Clustering of Stores by Geographic Distribution Across Urban, Suburban, and Rural Locations in Ecuador

The K-means clustering analysis with $k = 3$ is chosen, as this framework aligns with the urban, suburban, and rural framework, revealing how geographical context shapes consumer behavior and sales patterns across Ecuador.

The first cluster, visualized in red, is predominantly concentrated in Pichincha province, which includes the capital city of Quito. These stores likely represent urban metropolitan locations with higher population density and potentially greater affluence. This aligns with the regression results showing Quito as the baseline city with higher baseline sales compared to other regions, as evidenced by the negative coefficients for most other cities in Table 3. Urban stores demonstrate stronger performance in categories like GROCERY I and BEVERAGES, perhaps reflecting the greater population density in the urban areas and the consumption patterns of urban consumers.

The second cluster, shown in green, appears primarily in Guayas province and select other cities. This cluster represents stores in mid-sized cities or suburban areas surrounding major urban centers. The regression results support this distinction, with Guayas showing a significant negative coefficient compared to the Pichincha baseline, yet not as severe as more rural provinces. These suburban locations serve as intermediate markets between dense urban settings and rural communities, potentially explaining why promotional effectiveness varies across regions, as shown in the interaction terms of the regression model.

The third cluster, depicted in blue, demonstrates the greatest geographical diversity, spreading across multiple provinces including Santo Domingo de los Tsachilas, Manabi,

Azuay, and others. This rural distribution pattern aligns with the regression findings showing stronger negative coefficients for these regions, such as Manabi and Esmeraldas. The time series analysis further supports this clustering by revealing that the third lag of `payday_dummy` is significant, suggesting delayed spending effects that may be more pronounced in rural areas where consumers travel longer distances to shop and may consolidate their purchases after receiving income.

4.Evaluation and Conclusion

In conclusion, the different models yield several consistent findings. The fixed effects regression established that promotions significantly boost sales while oil prices negatively impact purchasing, with strong seasonal effects centered on December. The regularized regression models refined these insights, with Model 3 (the best model selected by AICc) highlighting the importance of location effects, product categories, and promotional timing, though it did not select payday variables as significant predictors. The deviance computed for the model 3 of lasso (57.19%) is almost identical to the R^2 in the fixed effect model (57.14%), which suggests strong consistency between different approaches, validating the robustness of the identified factors affecting sales.

The time series analysis complemented these findings by capturing temporal dependencies in sales patterns. While confirming the seasonal effects and promotional impacts, it uniquely identified a significant third lag of `payday_dummy`, suggesting delayed spending effects not captured by the regularization models. The K-means clustering provided geographical context to these patterns, showing how urban, suburban, and rural store locations experience different sales dynamics consistent with the coefficient patterns in our regression models.

Together, these models reveal that Favorita's sales are driven by a complex combination of promotional activities, regional characteristics, product categories, and economic conditions. The immediate impact of payday is less significant than anticipated, aligning with consumption smoothing theories, but with delayed effects particularly noticeable in rural areas. These insights suggest that Favorita should implement tailored strategies for different store clusters by optimizing seasonal promotional planning and monitoring economic indicators to maximize sales performance across Ecuador.